



**MACQUARIE**  
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**Macquarie University**

**Software Engineering (COMP2050)**

**ASSIGNMENT 1 — PROBLEM ANALYSIS**

**CHARGEMATE CUSTOMER**

**Software Requirements Specification**

**Group 3 – Team A**

**Team Members**

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Session 2 - 2026

# A1 Submission including Software Requirements Specification

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## SRS Structure

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Title Page

## ChargeMate Customer

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### Team A Members

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ChargeMate provides a seamless power bank rental service for visitors to Harbour North Plaza. Its aim is to reduce battery-related anxiety by allowing customers to easily find, rent, use and return portable power banks while keeping the experience clear, convenient and largely self-service.

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### Change Log

| Version | Date              | Changes                                                                                                                                                                                                        | Contributor(s) | Review / Agreement                                         |
|---------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------|
| 0.0     | 31 August 2026    | Initial project repository and supplied SRS/assignment structure established.                                                                                                                                  | Isaac          | Initial repository setup.                                  |
| 0.1     | 31 August 2026    | Created the initial meeting-minutes file structure for team and stakeholder records.                                                                                                                           | Isaac          | Repository record.                                         |
| 0.2     | 31 August 2026    | Added the ChargeMate Customer title and team members, initial vision statement, Purpose, Scope, and initial ChargeMate Customer/ChargeMate Ops boundaries.                                                     | Cooper         | Repository record.                                         |
| 0.3     | 7 September 2026  | Added the Overall Description, including Product Perspective, Product Functions, User Characteristics, Constraints, Assumptions and Dependencies, and the initial ChargeMate Customer context diagram.         | Yousef         | Merged into <a href="#">main</a> through PR #1.            |
| 0.4     | 7 September 2026  | Corrected SRS wording/typos and improved the initial document content.                                                                                                                                         | Cooper         | Repository record.                                         |
| 0.5     | 7 September 2026  | Added the SRS table of contents and improved document and meeting-minute formatting.                                                                                                                           | Harrison       | Repository record.                                         |
| 0.6     | 7 September 2026  | Added stakeholder clarification questions and responses concerning Customer–Ops information exchange, rental/return behaviour, pricing, guest rentals and error handling; added Team B clarification evidence. | Yousef         | Based on Team B stakeholder clarification.                 |
| 0.7     | 12 September 2026 | Added and refined functional and non-functional requirements with fit criteria, and updated the Customer–Ops context diagram and stakeholder clarification documentation.                                      | Yousef         | Merged into <a href="#">main</a> through PR #2.            |
| 0.8     | 12 September 2026 | Updated assumptions and dependencies and clarified pricing, rental-duration and late-fee responsibilities.                                                                                                     | Yousef         | Added to <a href="#">main</a> ; final team review pending. |

| Version | Date              | Changes                                                                                                                                                           | Contributor(s) | Review / Agreement                                |
|---------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------|
| 0.9     | 12 September 2026 | Added the complete ChargeMate Customer use case diagram and selected the three primary use cases for detailed analysis.                                           | Yousef         | Added to <b>main</b> ; final team review pending. |
| 1.0     | 12 September 2026 | Refined the vision statement, Purpose, Scope, ChargeMate Ops boundary, Product Perspective and Product Functions for consistency with the confirmed requirements. | Yousef         | Final team review pending.                        |
| 1.1     | 13 September 2026 | Added 1/3 of "important" Use cases. Elaborated on and added a description of Use Case 1 (Rent a Powerbank) and its Interaction Diagram.                           | Cooper         | Final team review pending.                        |
| 1.2     | 13 September 2026 | Added UC-02 Return a Power Bank description, alternative flows and sequence diagram.                                                                              | Harrison       | Reviewed and included in final submission.        |
| 1.3     | 13 September 2026 | Added UC-03 Handle Rental/Return Problem description, alternative flows and interaction diagram.                                                                  | Isaac          | Reviewed and included in final submission.        |
| 1.4     | 13 September 2026 | Completed the Group Activity Record discussion and outlook and added external service references.                                                                 | Cooper         | Reviewed and included in final submission.        |
| 1.5     | 13 September 2026 | Integrated stakeholder and meeting evidence into the appendices, completed the requirements traceability table, and performed final SRS consistency updates.      | Yousef         | Reviewed and included in final submission.        |

## Introduction

### Purpose

This document specifies the software requirements for ChargeMate Customer, the customer-facing component of the ChargeMate power bank rental service at Harbour North Plaza. It defines the system's functional and non-functional requirements, scope, users, constraints, dependencies, use cases and interactions. The SRS provides a requirements baseline for the subsequent design and development of ChargeMate Customer and its interaction with ChargeMate Ops.

# Scope

## ChargeMate Customer

ChargeMate Customer covers the customer-facing power bank rental experience at Harbour North Plaza.

The system will allow customers to:

- Find ChargeMate rental stations and view their current availability.
- View available power banks, charge status and available return slots.
- View pricing, rental-duration and applicable charge information.
- Start a power bank rental.
- View information about an active rental.
- Return a power bank to an operational ChargeMate station with an available return slot.
- Receive confirmation of successful rentals and returns.
- Receive clear information and support options when a rental, return or station-related problem occurs.
- Complete rentals using a guest or minimal-registration process.

ChargeMate Customer will exchange rental, return and station information with ChargeMate Ops.

## ChargeMate Ops

ChargeMate Ops is the operator-facing system responsible for managing the operational side of the ChargeMate network. Its responsibilities include station monitoring, power bank tracking, fault management, inventory and redistribution activities, and operational information required to run the service.

The detailed requirements for ChargeMate Ops are outside the scope of this SRS and are specified separately by Team B.

Staff operations, maintenance, station diagnostics, inventory redistribution and operator reporting are therefore outside the scope of ChargeMate Customer.

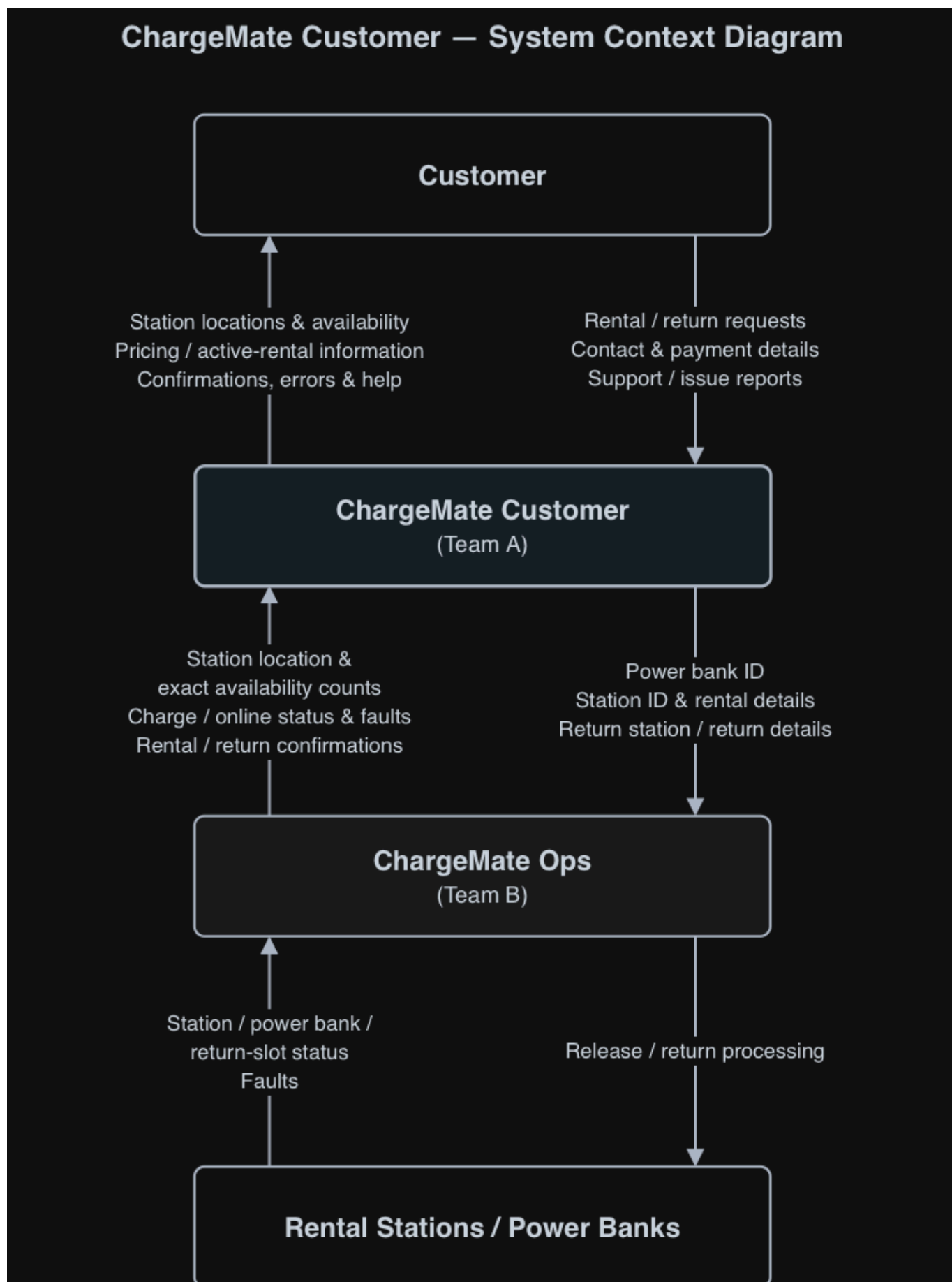
## Overall Description:

### Product Perspective

ChargeMate Customer is the customer-facing part of the ChargeMate power bank rental service at Harbour North Plaza. It is intended to help customers find, rent, use and return portable power banks with minimal assistance from shopping-centre staff.

The system operates alongside the physical rental stations, power banks and ChargeMate Ops system specified by Team B. ChargeMate Customer depends on ChargeMate Ops for station locations, power bank availability and charge status, return-slot availability, station status, and rental and return confirmations.

Staff operations, station maintenance, power bank redistribution and operator reporting are outside the scope of ChargeMate Customer.



**Figure 1: ChargeMate Customer System Context Diagram.** The diagram illustrates the system boundary and the high-level interactions between ChargeMate Customer, customers, the rental station network, and ChargeMate Ops.

## Product Functions

At a high level, ChargeMate Customer should allow customers to:

- Locate ChargeMate rental stations within Harbour North Plaza.
- Check whether a suitable power bank is available.
- View relevant rental and pricing information before beginning a rental.
- Start a power bank rental.
- View information about an active rental.
- Return a rented power bank.
- Receive confirmation when a rental or return has been completed successfully.
- Access clear help or support information if a problem occurs.

## User Characteristics

ChargeMate Customer may be used by a wide range of visitors to Harbour North Plaza, including shoppers, commuters, cinema visitors, library visitors, medical-centre visitors, and people attending weekend events.

Users may have different levels of technical experience and familiarity with the service. Some customers may be in a hurry, while others may only use ChargeMate once. Customers may also have concerns about payment, deposits, late fees, privacy, or what happens when a station does not operate correctly.

For this reason, the customer experience should aim to be clear, simple, and easy to understand without requiring extensive previous knowledge of the system.

## Constraints

**C-01:** ChargeMate Customer must remain focused on the customer-facing rental experience. Staff operations and maintenance functions are outside its scope.

**C-02:** The system should minimise the need for Harbour North Plaza customer-service staff to resolve routine technology or rental problems.

**C-03:** The rental process should avoid creating an unnecessarily large sign-up process for customers.

**C-04:** Some ChargeMate Customer functions depend on information provided by the physical rental stations and ChargeMate Ops.

## Assumptions and Dependencies

**A-01:** ChargeMate Customer depends on ChargeMate Ops for station information, including station location, exact power bank availability, power bank charge status, available return slots and station operational status.

**A-02:** ChargeMate Customer depends on ChargeMate Ops to confirm that a rental has successfully started and to identify the power bank released to the customer.

**A-03:** ChargeMate Customer depends on ChargeMate Ops to confirm that a returned power bank has been successfully recorded.

**A-04:** ChargeMate Customer manages and displays customer-facing pricing, deposit/pre-authorisation, rental-duration and late-fee information, including determining when a rental becomes overdue and calculating applicable late fees.

**A-05:** When starting a rental, ChargeMate Customer provides ChargeMate Ops with the selected power bank ID, station ID and the rental details required to record the rental.

## Specific Requirements

### Functional Requirements

#### Station Information and Availability

##### **FR-01: Station Location**

The system shall display the location of each ChargeMate rental station within Harbour North Plaza.

##### **Fit Criterion:**

When station information is available, the customer can view the location of every operational ChargeMate station.

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##### **FR-02: Power Bank Availability**

The system shall display the exact number of available power banks at each ChargeMate station.

##### **Fit Criterion:**

For each station, the number displayed by ChargeMate Customer matches the availability information supplied by ChargeMate Ops.

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##### **FR-03: Power Bank Charge Status**

The system shall indicate whether available power banks are sufficiently charged for rental.

##### **Fit Criterion:**

A power bank identified by ChargeMate Ops as sufficiently charged is shown as available for rental, while an insufficiently charged power bank is not shown as available.

---

##### **FR-04: Return Slot Availability**

The system shall display the exact number of available return slots at each ChargeMate station.

##### **Fit Criterion:**

For each station, the number of available return slots displayed to the customer matches the latest information supplied by ChargeMate Ops.

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##### **FR-05: Station Status**

The system shall display whether a ChargeMate station is operational or unavailable.

##### **Fit Criterion:**

When ChargeMate Ops reports a station as online, offline or unavailable, ChargeMate Customer displays the corresponding station status.

---

**FR-06: Availability Updates**

The system shall update station and power bank availability whenever a rental, return, fault or station-status change occurs.

**Fit Criterion:**

After ChargeMate Customer receives an updated availability or status event from ChargeMate Ops, the displayed station information reflects the updated values.

**Rental Process****FR-07: Start Rental**

The system shall allow a customer to start a rental for an available power bank.

**Fit Criterion:**

When a customer starts a valid rental, ChargeMate Customer sends the selected power bank ID, station ID and required rental details to ChargeMate Ops.

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**FR-08: Rental Confirmation**

The system shall confirm to the customer when a rental has successfully started.

**Fit Criterion:**

After ChargeMate Ops confirms a successful rental, ChargeMate Customer displays confirmation and identifies the released power bank.

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**FR-09: Failed Power Bank Release**

The system shall inform the customer when a rental is authorised but the station fails to release the power bank.

**Fit Criterion:**

When a release failure is reported, the system displays a failure message and the unsuccessful rental or payment authorisation is cancelled or reversed.

**Return Process****FR-10: Cross-Station Return**

The system shall allow customers to return a rented power bank to any operational ChargeMate station with an available return slot.

**Fit Criterion:**

A customer can select and complete a return at a station different from the original rental station when that station is operational and has at least one available return slot.

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**FR-11: Record Return**

The system shall provide the power bank ID and return station information when recording a return.

**Fit Criterion:**

When a return is processed, ChargeMate Ops receives the correct power bank ID and return-station information.

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**FR-12: Return Confirmation**

The system shall notify the customer when a return has been successfully recorded.

**Fit Criterion:**

After ChargeMate Ops confirms the return, ChargeMate Customer displays a successful return confirmation.

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**FR-13: Full Return Station**

The system shall inform the customer when a selected station has no available return slots and provide another nearby station with available capacity.

**Fit Criterion:**

When the selected station has zero available return slots, the system prevents the return from being directed to that station and displays at least one nearby operational station with an available return slot, where one exists.

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**FR-14: Unconfirmed Return**

The system shall inform the customer when a physical return cannot be confirmed.

**Fit Criterion:**

When the return is not confirmed, the system displays an error message and provides instructions for contacting support or reporting the issue.

**Customer and Active Rental Information****FR-15: Guest Rental**

The system shall support guest or minimal-registration rentals.

**Fit Criterion:**

A customer can begin a rental without creating a full customer account and is only required to provide the necessary contact and payment information.

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**FR-16: Active Rental Information**

The system shall display relevant information about the customer's active rental.

**Fit Criterion:**

During an active rental, the customer can view:

- Rental start time.
- Elapsed rental time.
- Current price.
- Rental status.
- Available return locations.

**Pricing and Rental Charges**

**FR-17: Pricing and Rental Charges**

The system shall manage and display pricing, deposit/pre-authorisation, rental-duration and applicable late-fee information to the customer.

**Fit Criterion:**

Before and during a rental, the customer can view the applicable pricing and rental-duration information. When a rental becomes overdue, the system identifies the overdue status and calculates and displays any applicable late fee.

**Error Handling****FR-18: Customer Error Messages**

The system shall display clear messages when a rental or station-related problem occurs.

**Fit Criterion:**

The system displays an appropriate customer-facing message when any of the following occurs:

- A station is unavailable.
- A power bank is unavailable.
- No return slots are available.
- A power bank release fails.
- A return cannot be confirmed.
- A service error occurs.

**Non-Functional Requirements****NFR-01: Usability**

The system shall provide a clear and simple customer experience suitable for first-time and occasional users.

**Fit Criterion:**

A customer shall be able to identify station availability, rental information, active rental status and return options without requiring assistance from Harbour North Plaza staff.

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**NFR-02: Self-Service Support**

The system shall provide customers with sufficient information to resolve common rental and return problems without relying on shopping-centre staff.

**Fit Criterion:**

For supported error conditions, including unavailable stations, failed releases, unavailable return slots and unconfirmed returns, the system displays a clear explanation and the appropriate next action or support option.

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**NFR-03: Information Consistency**

Customer-facing station and rental information shall remain consistent with information received from ChargeMate Ops.

**Fit Criterion:**

When ChargeMate Ops provides updated station availability, return-slot availability or station status, ChargeMate Customer displays values that match the latest received information.

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**NFR-04: Privacy**

The system shall minimise the personal information required from customers when using the rental service.

**Fit Criterion:**

A guest or minimal-registration rental can be completed using only the contact and payment information required to process and manage the rental.

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**NFR-05: Error Clarity**

Customer-facing error messages shall clearly communicate the problem and, where possible, what the customer should do next.

**Fit Criterion:**

For each defined customer-facing error condition, the displayed message identifies the problem and provides either a recovery action, an alternative option or instructions for contacting support.

**Requirements Traceability**

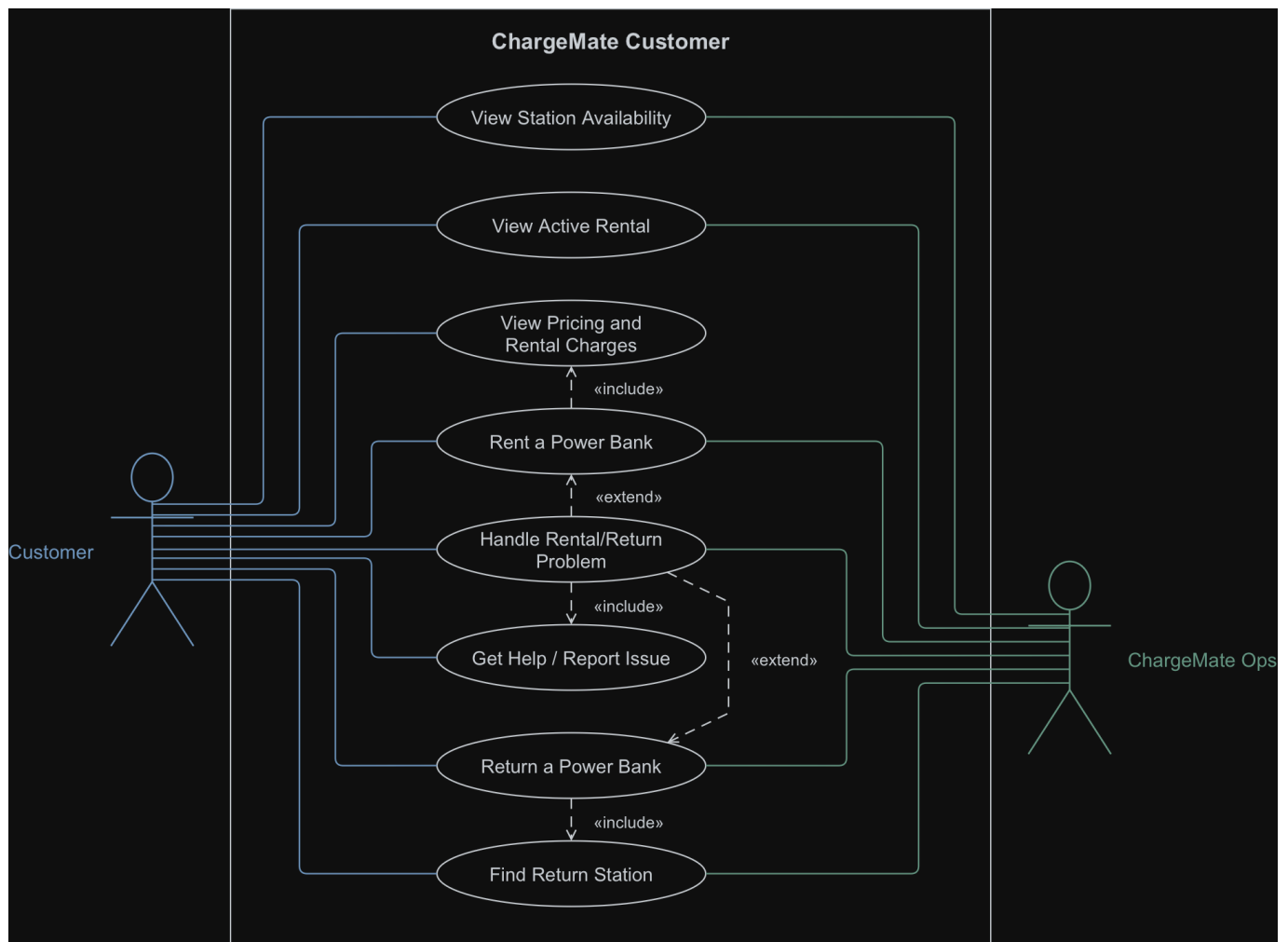
The following table traces the functional and non-functional requirements to the relevant use cases, system models and stakeholder evidence. This demonstrates how the requirements identified during elicitation are represented throughout the SRS.

| Requirement                           | Use Case / Model Coverage                                                  | Supporting Evidence |
|---------------------------------------|----------------------------------------------------------------------------|---------------------|
| <b>FR-01 Station Location</b>         | View Station Availability; Find Return Station; UC-02                      | Appendix A, Q1      |
| <b>FR-02 Power Bank Availability</b>  | View Station Availability; Rent a Power Bank; UC-01                        | Appendix A, Q1–Q2   |
| <b>FR-03 Power Bank Charge Status</b> | View Station Availability; Rent a Power Bank; UC-01                        | Appendix A, Q1      |
| <b>FR-04 Return Slot Availability</b> | View Station Availability; Find Return Station; Return a Power Bank; UC-02 | Appendix A, Q1–Q2   |
| <b>FR-05 Station Status</b>           | View Station Availability; Find Return Station; UC-02                      | Appendix A, Q1      |
| <b>FR-06 Availability Updates</b>     | View Station Availability; Context Diagram                                 | Appendix A, Q3      |
| <b>FR-07 Start Rental</b>             | Rent a Power Bank; UC-01; UC-01 Interaction Diagram                        | Appendix A, Q4      |
| <b>FR-08 Rental Confirmation</b>      | Rent a Power Bank; UC-01; UC-01 Interaction Diagram                        | Appendix A, Q5      |

| Requirement                             | Use Case / Model Coverage                                                          | Supporting Evidence                 |
|-----------------------------------------|------------------------------------------------------------------------------------|-------------------------------------|
| <b>FR-09 Failed Power Bank Release</b>  | Rent a Power Bank; Handle Rental/Return Problem; UC-01; UC-03                      | Appendix A, Q9                      |
| <b>FR-10 Cross-Station Return</b>       | Return a Power Bank; Find Return Station; UC-02                                    | Appendix A, Q7                      |
| <b>FR-11 Record Return</b>              | Return a Power Bank; UC-02; UC-02 Interaction Diagram                              | Appendix A, Q6                      |
| <b>FR-12 Return Confirmation</b>        | Return a Power Bank; UC-02; UC-02 Interaction Diagram                              | Appendix A, Q6                      |
| <b>FR-13 Full Return Station</b>        | Find Return Station; Return a Power Bank; UC-02                                    | Appendix A, Q8                      |
| <b>FR-14 Unconfirmed Return</b>         | Return a Power Bank; Handle Rental/Return Problem; UC-02; UC-03                    | Appendix A, Q10                     |
| <b>FR-15 Guest Rental</b>               | Rent a Power Bank; UC-01                                                           | Appendix A, Q13                     |
| <b>FR-16 Active Rental Information</b>  | View Active Rental                                                                 | Appendix A, Q14                     |
| <b>FR-17 Pricing and Rental Charges</b> | View Pricing and Rental Charges; Rent a Power Bank; View Active Rental; UC-01      | Appendix A, Q11–Q12                 |
| <b>FR-18 Customer Error Messages</b>    | Handle Rental/Return Problem; Get Help / Report Issue; UC-03                       | Appendix A, Q15                     |
| <b>NFR-01 Usability</b>                 | Applies across all customer-facing use cases                                       | Project Scope; User Characteristics |
| <b>NFR-02 Self-Service Support</b>      | Handle Rental/Return Problem; Get Help / Report Issue; UC-03                       | Appendix A, Q15                     |
| <b>NFR-03 Information Consistency</b>   | View Station Availability; Rent a Power Bank; Return a Power Bank; Context Diagram | Appendix A, Q1–Q3                   |
| <b>NFR-04 Privacy</b>                   | Rent a Power Bank; UC-01                                                           | Appendix A, Q13                     |
| <b>NFR-05 Error Clarity</b>             | Handle Rental/Return Problem; Get Help / Report Issue; UC-03                       | Appendix A, Q15                     |

Use cases and interactions

## Use Case Diagram



**Figure 2: ChargeMate Customer Use Case Diagram.** The diagram shows the main interactions between the Customer, ChargeMate Customer system and ChargeMate Ops. The three primary use cases selected for detailed analysis are Rent a Power Bank, Return a Power Bank, and Handle Rental/Return Problem.

## UC-01: Rent a Power Bank

| Field                 | Details                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case              | UC-01: Rent a Power Bank                                                                                                                                                                                                                                                                                                                                                     |
| Goal                  | The customer successfully rents an available, fully charged power bank from a ChargeMate station for use during their visit to Harbour North Plaza.                                                                                                                                                                                                                          |
| Preconditions         | <ul style="list-style-type: none"> <li>- At least one ChargeMate station is operational and reporting availability information.</li> <li>- At least one power bank at the selected station is available and adequately charged (FR-03).</li> <li>- The customer can reach the ChargeMate Customer UI and provide the contact/payment details required for rental.</li> </ul> |
| Success End Condition | The power bank is unlocked for the customer, ChargeMate Ops has recorded the rental against the power bank ID, station ID and rental/payment details (A-05); the customer's current rental information becomes available to view (FR-16).                                                                                                                                    |

| Field                       | Details                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Failed End Condition</b> | No power bank is unlocked and no rental record exists in ChargeMate Ops, any payment made for the attempt is cancelled or refunded (FR-09). |
| <b>Primary Actors</b>       | Customer                                                                                                                                    |
| <b>Secondary Actors</b>     | ChargeMate Ops                                                                                                                              |
| <b>Trigger</b>              | Customer selects a fully charged, available power bank at a ChargeMate station and chooses to begin a rental.                               |

### Description / Main Success Scenario

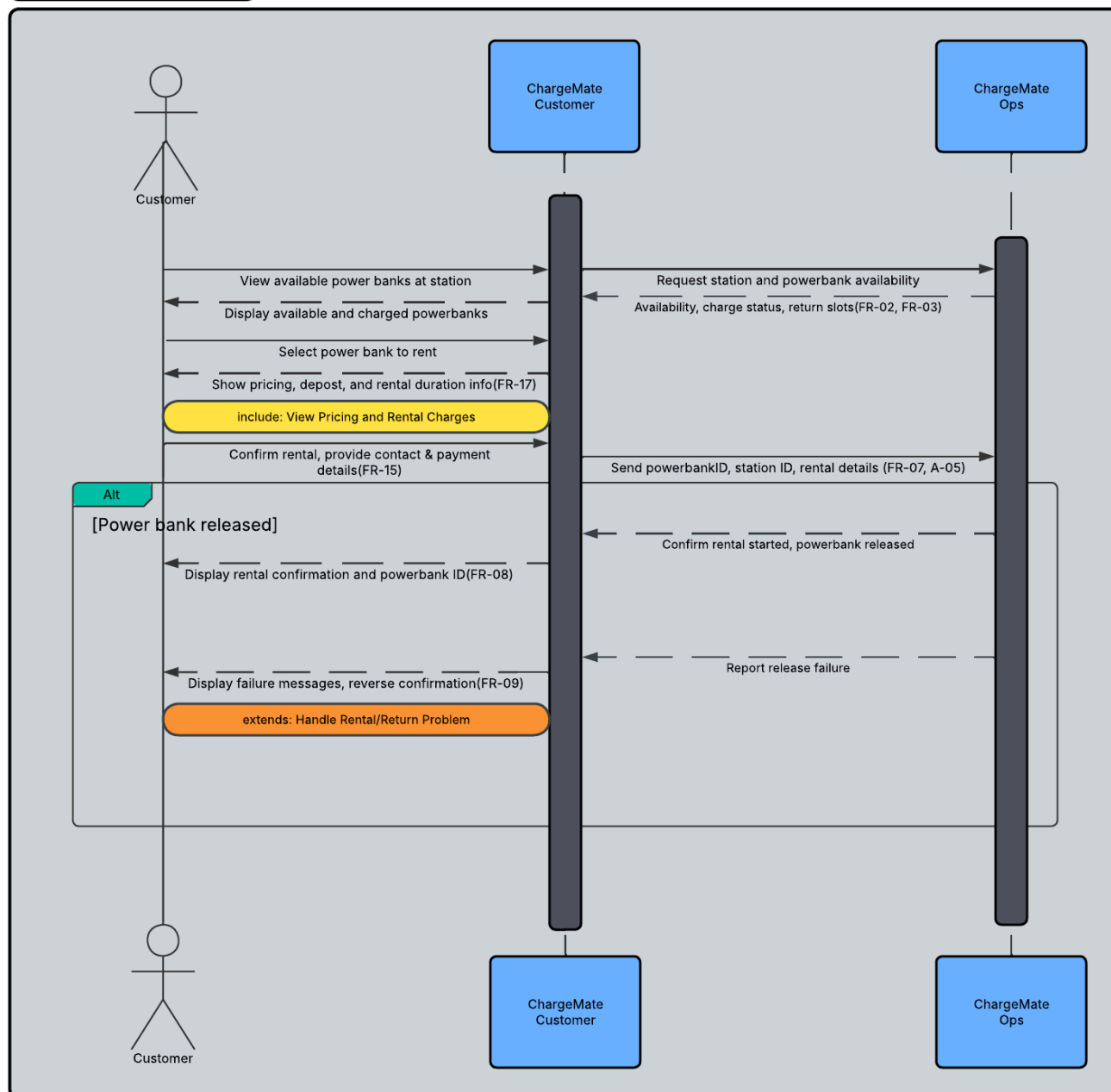
| Step | Action                                                                                                                      |
|------|-----------------------------------------------------------------------------------------------------------------------------|
| 1    | Customer views the available, adequately charged power banks at a ChargeMate station (FR-02, FR-03).                        |
| 2    | Customer selects a power bank and requests to rent it.                                                                      |
| 3    | System displays pricing, deposit and rental duration information (FR-17) - <i>include: View Pricing and Rental Charges.</i> |
| 4    | Customer confirms the rental and provides the contact and payment details required for a rental (FR-15).                    |
| 5    | System sends the power bank ID, station ID and rental/payment details to ChargeMate Ops (FR-07, A-05).                      |
| 6    | ChargeMate Ops authorises the rental and unlocks the power bank at the station.                                             |
| 7    | System displays a rental confirmation in conjunction with the released power bank (FR-08).                                  |
| 8    | Use case ends, the rental becomes active.                                                                                   |

### Alternative Flows

| Step | Branching Action                                                                                                                                                                                                                                                                                              |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1a   | No power banks are available at the selected station: System displays an unavailable message (FR-18) and, where possible, displays a nearby station with availability.                                                                                                                                        |
| 4a   | Contact or payment details are invalid: System displays an error and returns the customer to step 4 to re-enter details.                                                                                                                                                                                      |
| 6a   | <b>Release Failure:</b> Ops reports that the rental was authorised but the station failed to unlock the power bank (FR-09). System displays a failure message, cancels/refunds payment type, and the use case extends into <i>Handle Rental/Return Problem</i> so the customer can get further help (NFR-02). |

## Interaction Diagram - Rent a Power Bank

## Interaction Diagram



## UC-02 - Returning a Power Bank

| Field                 | Details                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case              | UC-02: Returning a Power Bank                                                                                                                            |
| Goal                  | Allow the client to return the used power bank to any station within Harbour North Plaza that has an available slot.                                     |
| Preconditions         | - The Client has atleast 1 powerbank that they want to return, and there is an available slot in a working station (FR-04) (FR-05)in Harbour North Plaza |
| Success End Condition | The Power bank is return to the station; Power bank detail are successfully added back to the system                                                     |

| Field                       | Details                                                           |
|-----------------------------|-------------------------------------------------------------------|
| <b>Failed End Condition</b> | No slot is available for return, power bank is faulty upon return |
| <b>Primary Actors</b>       | Customer                                                          |
| <b>Secondary Actors</b>     | ChargeMate Ops                                                    |
| <b>Trigger</b>              | The Client wants to return a power bank                           |

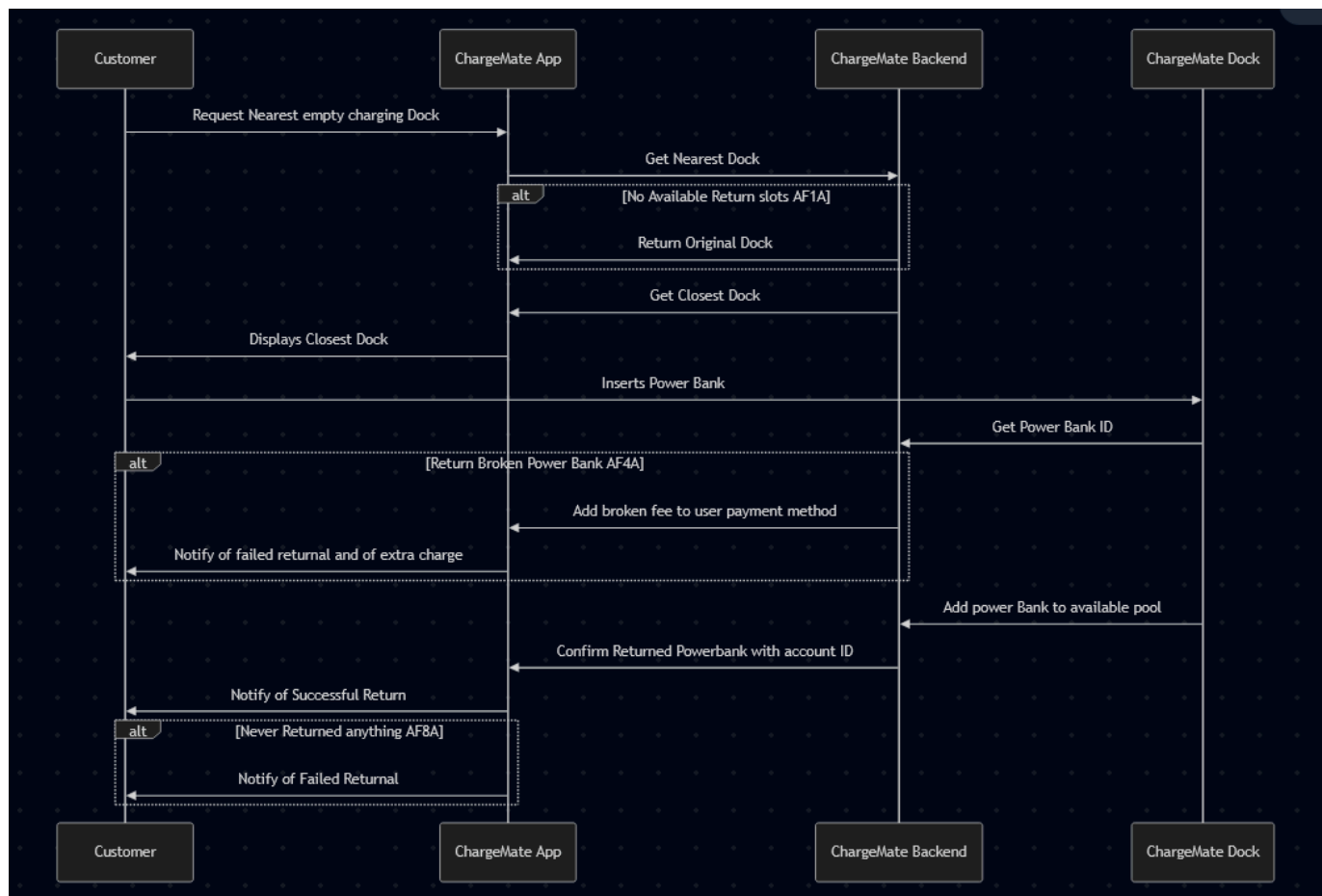
### Description / Main Success Scenario

| Step | Action                                                                                                             |
|------|--------------------------------------------------------------------------------------------------------------------|
| 1    | The user access the Charge mate app and uses the find station function (FR-01)                                     |
| 2    | The system retrieves information about which power stations have slots and if they are functioning (FR-05) (FR-06) |
| 3    | The system displays the nearest station with an empty slot (FR-01)                                                 |
| 4    | The user inserts the power bank back into the empty slot of the nearby station                                     |
| 5    | The system receives the power bank ID and return station information (FR-11)                                       |
| 6    | System adds the recieved power bank back into the pool of available power banks (FR-06)                            |
| 7    | The system notifies the client of successful return (FR-12)                                                        |
| 8    | Use case ends, power bank becomes available to rent again                                                          |

### Alternative Flows

| Step | Branching Action                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1a   | the nearest available station has no available return slots, the system prompts the user to return to the original station using the station ID |
| 4a   | Power bank is damaged on return, send a notification to the customer to contact customer support and incur a damaged goods fee.                 |
| 8a   | If a return is not completed, the system notifies the user to contact the customer support                                                      |

### Sequence Diagram



## UC-03: Handle Rental/Return Problem

| Field                 | Details                                                                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case              | UC-03: Handle Rental/Return Problem                                                                                                                                                                                                 |
| Goal                  | Provide clear self-service instructions, error handling, and support escalation paths when a rental, return, or station fault occurs. (NFR-02, NFR-05).                                                                             |
| Preconditions         | <ul style="list-style-type: none"> <li>- A rental/return fault has occurred (FR-09, FR-14).</li> <li>- The customer can reach the ChargeMate Customer UI or manually select "Report Problem" on an active rental screen.</li> </ul> |
| Success End Condition | Issue details are logged with ChargeMate Ops, automated fee adjustments or holds are applied and clear guidance or support options are presented to the customer (NFR-02, NFR-05).                                                  |
| Failed End Condition  | System cannot process fault report automatically; customer is provided direct support contact credentials (FR-18).                                                                                                                  |
| Primary Actors        | Customer                                                                                                                                                                                                                            |
| Secondary Actors      | ChargeMate Ops                                                                                                                                                                                                                      |
| Trigger               | System detects a hardware/network exception (failed release or unconfirmed return) or customer selects "Report Issue" in ChargeMate Cutomer (FR-18).                                                                                |

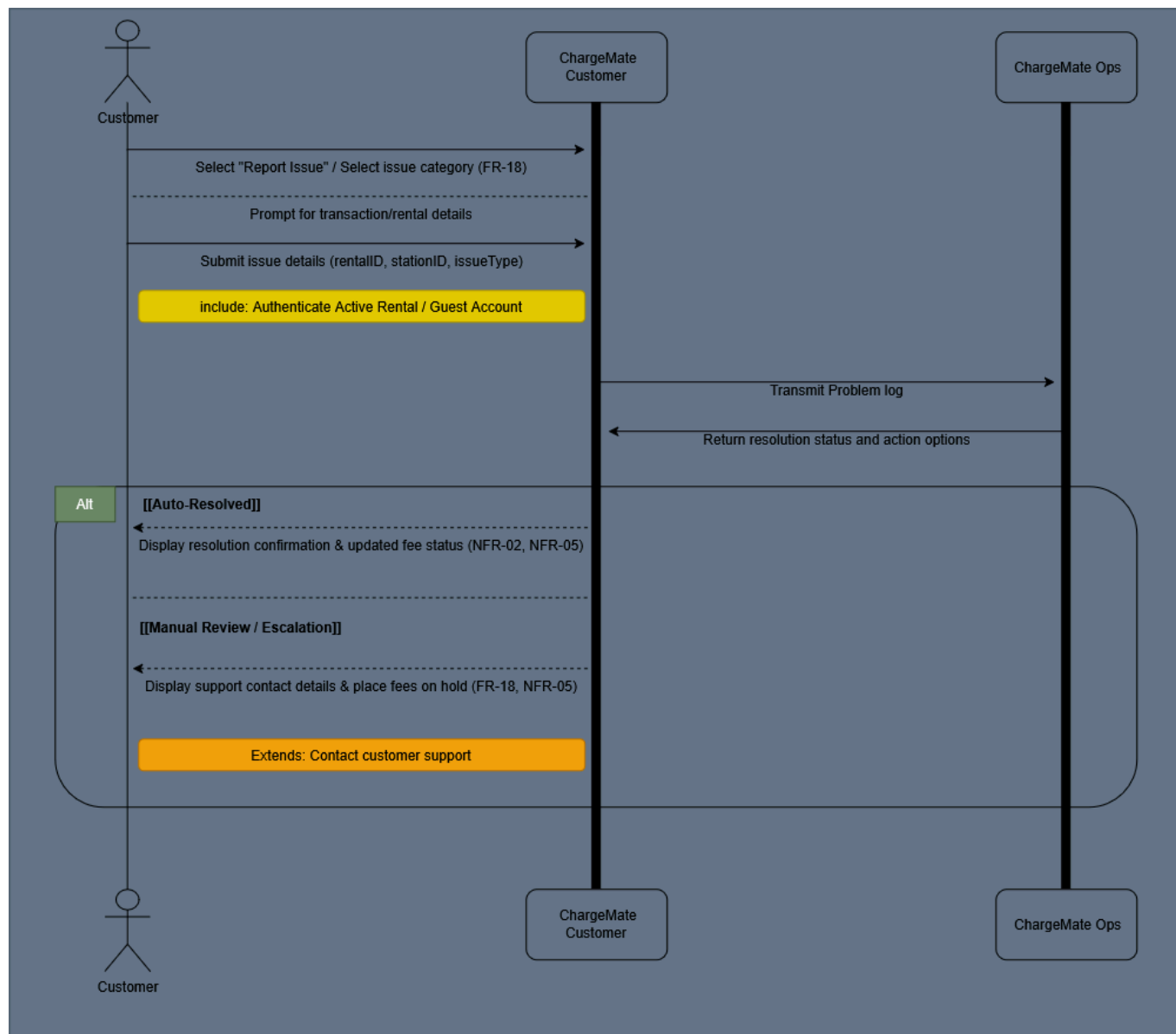
## Description/ Main Success Scenario

| Step | Action                                                                                                                      |
|------|-----------------------------------------------------------------------------------------------------------------------------|
| 1    | Customer selects "Help / Report Issue" or system auto-redirects from a failed transaction (FR-18).                          |
| 2    | System displays issue categories (e.g., Unconfirmed Return, Failed Release, Damaged Bank).                                  |
| 3    | Customer selects an issue category and submits transaction details.                                                         |
| 4    | System captures transaction context (Rental ID, Station ID, Power Bank ID) and transmits problem log to ChargeMate Ops.     |
| 5    | ChargeMate Ops processes report and responds with resolution options or automated actions (e.g., reverse charge, hold fee). |
| 6    | System displays clear next steps or resolution confirmation to the customer (NFR-02, NFR-05).                               |

## Alternative Flows

| Step | Branching Action                                                                                                                                                                                                                                          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1a   | <b>Unconfirmed Physical Return (FR-14):</b> Station fails to acknowledge power bank insertion within timeout. System auto-redirects to problem reporting, captures station ID, places rental on temporary hold, and displays self-service steps (NFR-02). |
| 1b   | <b>Failed Power Bank Release (FR-09):</b> Station lock fails during unlock. System auto-initiates payment authorization reversal/refund, logs station fault with ChargeMate Ops, and displays confirmation of cancellation to customer (FR-18).           |
| 3a   | <b>Customer Disputes Overdue / Late Fee (FR-17):</b> Customer reports issue regarding incorrect late fee. System logs time discrepancy report with ChargeMate Ops and pauses pending collection while under review.                                       |
| 4a   | <b>Network Connection / Service Timeout:</b> System is unable to reach ChargeMate Ops to submit problem log. System displays offline error message with direct customer support contact phone number / QR code (FR-18, NFR-05).                           |
| 5a   | <b>Automated Resolution Unavailable:</b> ChargeMate Ops flags issue for manual review. System places pending late fees on temporary hold and displays support contact instructions (NFR-02, NFR-05).                                                      |

## Interaction Diagram - Handle Rental/Return Problem



## Group Activity Record

### Discussion

#### Elicitation Methods:

Our team used three elicitation techniques, each one contributing a different perspective and idea to our requirements.

- **Document Analysis:** We began by analysing the ChargeMate Customer project brief supplied to Team A. We broke down the information supplied in the project brief to contextualise and grasp the assigned task. The client's concerns involve avoiding a long sign-up requirement, and to minimise the need for employees and human interaction. The distinction between the project scope of ChargeMate Customer and ChargeMate Ops was an extremely important task to prevent any confusion between the teams. Our interpretation of the given project brief allowed us to extract any relevant specifications and requirements before we put anything directly to Team B.
- **Written Q&A with our stakeholder** If the project brief analysis surfaced any specific uncertainties, we drafted a structured set of written clarification questions and sent them to our acting

Stakeholders (Team B). These covered the type of information exchanged with ChargeMate Ops, the exact count vs an available/unavailable status, cross station returns, actions to take in case of failure, pricing and penalties, registration, and active rental information. (see Appendix A: Stakeholder Interaction Log). Their answers shaped ~15 of our functional and non functional requirements, with two in particular that we would have gotten wrong, those involving pricing/penalties and the guest rental scope.

- **Analysis & Comparison of existing power bank rental services** To cross check our design choices against how similar problems exist and are solved in the a professional and commercial environment, we looked at two existing power bank rental services: LifeSaver Power (UK) and Ezycharge (Australia). Lifesaver never charges returned power banks inside the machine, instead powerbanks are charged and cleaned off-site, then restocked. This is completely different to our model, since our brief requires the service to run with minimal human involvement, which rules out the possibility of any similar collection feature. Although the possibility of a routine collection at stations have been ruled out, this flagged hygiene as a requirement gap that's worth further investigation. Ezycharge shares a similar nation-wide network, where customers can return a power bank to any active Ezycharge kiosk location regardless of where they originally rented, similar to our cross station return requirement (FR-10). It validated that the task wasn't unrealistic, and shows it is a plausible and convenient feature.

## Outlook:

If we were to continue this project beyond our current SRS, there would be several areas that need more clarification and research before anything is in its final implementation stage.

- Firstly, our requirements are drafted in a written Q&A format with Team B as a stakeholder, not real ChargeMate customers or individuals who will actually interact with the service. Any assumption that we have would have to be validated like how customers would react to any failures and how they would be resolved. A prototype could be tested with real customers such as shoppers, employees, etc. It would provide information and statistics on the average duration of rental, spikes in demand, and more.
- Secondly, our look at Lifesaver revealed a gap that we overlooked: hygiene. Our brief doesn't currently address how a shared and touched powerbank is kept clean between rentals, and this is a concern worth raising with ChargeMate rather than assuming. A simple option that can be explored would be built in UV sanitisation lights in rental stations that would operate between rentals. Before turning hygiene into a requirement, clarification from the client side would be necessary to confirm if it is a concern worth specifying.
- Thirdly, a few requirements rely on assumptions that we've taken from Team B's answers rather than confirming with the actual implementation of ChargeMate Ops. For example, updating when a change occurs like an unlocked port (FR-06) says "whenever a change occurs", not how quickly, and what ChargeMate Customer should do if Ops is unavailable when a customer tries to rent.
- Finally, business details and specifics like pricing Charts, deposit amounts, late fees, and payment types are currently unspecified. These were not our team's responsibility to define, but the final design will need exact numbers.

To make sure we have confidence in our requirements, a follow up meeting with Team B discussing our SRS would be significantly helpful, as we can contrast requirements, clarify and validate any assumptions, and flag anything that doesn't match their design choices.

## Individual Contributions

- Outline what each team member has led, contributed to, discussed, and/or reviewed.

| Member   | Contributions                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yousef   | documented the confirmed responses from Team B. Developed the Overall Description and Customer Ops context diagram. Created and refined FR-01 to FR-18 and NFR-01 to NFR-05. Updated the assumptions and dependencies, including pricing, rental duration, late fees, guest rentals and Customer–Ops responsibilities. Created the overall ChargeMate Customer use case diagram and selected the three main use cases for detailed analysis. |
| Harrison | UC-02 created description, goals, scenarios, Steps and alternative flows. UC-02 Sequence diagram Diagrams, Reviewed Fit Criteria and Functional Requirements, Worked on the Appendix for external tools and worked on the Individual Contributions section                                                                                                                                                                                   |
| Cooper   | chose use case one - Rent a powerbank as 1/3 of the most important use cases, created a description and full use case details, goals, scenarios and interaction/sequence diagram for UC1. Wrote the discussion responses for the teams elicitation methods and wrote down the teams future outlook. Added appendix A to srs.md. Drafted the initial project scope and purpose.                                                               |
| Isaac    | UC-03 created description, goals, scenarios, Steps and alternative flows. UC-03 Sequence diagram Diagrams. Refined the project scope, purpose, vision and system boundaries for consistency and maintained the SRS change log through the requirements and modelling stages. Formatting on Minutes                                                                                                                                           |

## Appendices

### Appendix A — Stakeholder Interaction Log

The following stakeholder clarification was used to refine the requirements for ChargeMate Customer.

| Interaction                                   | Stakeholder | Purpose                                                                                                | Main Outcomes                                                                                                                                                    | Requirements Affected                          |
|-----------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Requirements clarification — 7 September 2026 | Team B      | Clarify information exchange, rental and return behaviour, pricing responsibilities and error handling | Confirmed exact availability counts, cross-station returns, guest rentals, failed-release behaviour, active-rental information and Customer/Ops responsibilities | FR-01–FR-18, NFR-02, NFR-04, NFR-05, A-01–A-05 |

| #   | Question                                                                                                                                                                        | Team B's Answer                                                                                                               | Requirements Affected                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1.  | What information will ChargeMate Customer receive from ChargeMate Ops about each rental station (location, available power banks, charge status, return slots, online/offline)? | All five confirmed: station location, exact power bank availability, charge status, return-slot availability, station status. | FR-01, FR-02, FR-03, FR-04, FR-05, A-01 |
| 2.  | Should availability be shown as exact numbers or a simple available/unavailable status?                                                                                         | Exact numbers.                                                                                                                | FR-02, FR-04                            |
| 3.  | How frequently will station and power bank availability be updated?                                                                                                             | Whenever a rental, return, fault, or station-status change occurs.                                                            | FR-06                                   |
| 4.  | When a customer starts a rental, what does Customer need to send to Ops?                                                                                                        | Power bank ID, station ID, and the rental details needed to record the rental.                                                | FR-07, A-05                             |
| 5.  | What does Ops send back to confirm a rental has started?                                                                                                                        | Confirmation that the rental started successfully and identification of the released power bank.                              | FR-08, A-02                             |
| 6.  | What does Customer send Ops on return, and what confirmation comes back?                                                                                                        | Power bank ID and return station sent; confirmation returned once the return is successfully recorded.                        | FR-11, FR-12, A-03                      |
| 7.  | Can a customer return to any station, or only the original rental station?                                                                                                      | Any operational ChargeMate station with an available return slot.                                                             | FR-10                                   |
| 8.  | What happens if the selected return station has no available slots?                                                                                                             | Inform the customer and show another nearby station with available return slots.                                              | FR-13                                   |
| 9.  | What happens if payment/authorisation succeeds but the station fails to release the power bank?                                                                                 | Inform the customer of the release failure and cancel/reverse the unsuccessful rental or payment authorisation.               | FR-09                                   |
| 10. | What happens if the customer physically returns the power bank but the system doesn't confirm it?                                                                               | Inform the customer the return couldn't be confirmed and provide instructions to contact support/report the issue.            | FR-14                                   |

| #   | Question                                                                            | Team B's Answer                                                                                               | Requirements Affected |
|-----|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------|
| 11. | Which system determines overdue rentals and calculates late fees — Customer or Ops? | Customer system determines overdue status and calculates applicable late fees.                                | FR-17, A-04           |
| 12. | Does Ops manage the pricing/deposit/late-fee information Customer displays?         | No, Customer manages and displays pricing, deposits/authorisation, late fees and rental duration information. | FR-17, A-04           |
| 13. | Does Ops require a full account, or can rentals be completed as a guest?            | Guest/minimal-registration rentals allowed; only necessary contact and payment details required.              | FR-15, NFR-04         |
| 14. | What active-rental information should Customer be able to retrieve?                 | Start time, elapsed time, current price, rental status, and available return locations.                       | FR-16                 |
| 15. | Are there Ops error/status messages Customer must specifically handle?              | Yes, unavailable stations/power banks, no return slots, failed release/return, and service errors.            | FR-18, NFR-02, NFR-05 |

## Appendix B — Team and Stakeholder Meeting Minutes

The following meeting records document the team's requirements analysis, stakeholder clarification, project coordination and SRS development activities throughout Assignment 1.

| Date              | Meeting Focus                                                       |
|-------------------|---------------------------------------------------------------------|
| 11 August 2026    | Team setup, project allocation and initial requirements planning    |
| 18 August 2026    | Requirements clarification and primary use-case planning            |
| 24 August 2026    | Interaction-diagram planning and Customer–Ops responsibility review |
| 1 September 2026  | SRS baseline development and requirements refinement                |
| 7 September 2026  | Team B stakeholder requirements clarification                       |
| 12 September 2026 | Final requirements, fit criteria, use cases and SRS review          |

### Meeting Minutes — 11 August 2026

**Date:** 11 August 2026

**Meeting Place:** Macquarie University Library

**Present:** Harrison McLuckie, Cooper Went, Isaac Lynn, Yousef Mustafa

**Apologies:** None

**Minute Taker:** Isaac Lynn

## Agenda

1. Establish team communication channels and schedule.
2. Confirm project selection — Team A on Project A, ChargeMate Customer, with Team B representing ChargeMate Ops.
3. Review preliminary specification scope and draft initial Q&As.

## Discussion Summary

- Team A and Team B established joint WhatsApp communication channels.
- Team A's assignment to ChargeMate Customer was confirmed, with Team B serving as the client team representing ChargeMate Ops.
- The team initiated a review of the system boundary between ChargeMate Customer and ChargeMate Ops.
- Information gaps were identified regarding station availability, hardware-release failures, guest-rental constraints and return validation.

## Decisions

- Communication will occur through WhatsApp, while the SRS and project files will be maintained through GitHub.
- The team agreed to develop formal questions covering system inputs, outputs and exception flows before the next practical.

## Actions

- Draft the initial list of questions for Team B.
- Review initial FR/NFR mappings for the core rental and return flows.

## Next Meeting

18 August 2026

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## Meeting Minutes — 18 August 2026

**Date:** 18 August 2026

**Meeting Place:** Zoom

**Present:** Harrison McLuckie, Cooper Went, Isaac Lynn, Yousef Mustafa

**Apologies:** None

**Minute Taker:** Isaac Lynn

## Agenda

1. Review the initial draft of Team A's interview questions with the Client.
2. Refine the scope of UC-01 — Rent a Power Bank and UC-02 — Return a Power Bank.
3. Determine delegation of SRS diagramming and technical boundary sections.

## Discussion Summary

- Team A presented initial questions concerning real-time status updates, error handling and actor responsibilities.
- Edge cases were discussed, including station release failures (FR-09) and unconfirmed physical returns (FR-14).
- Guest/minimal-registration requirements were discussed.
- UC-01, UC-02 and UC-03 were selected as the three primary use cases.

## Decisions

- Team A would finalise and submit the detailed technical clarification questions to the Client.
- UC-01 diagramming was allocated to Cooper, UC-02 to Harrison and UC-03 to Isaac.

## Actions

- Draft the interaction/sequence diagram structure for UC-01.
- Draft the interaction/sequence diagram structure for UC-02.
- Draft the initial use-case specification for UC-03.

## Next Meeting

24 August 2026

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## Meeting Minutes — 24 August 2026

**Date:** 24 August 2026

**Meeting Place:** 4RPD, Room 113

**Present:** Harrison McLuckie, Cooper Went, Isaac Lynn, Yousef Mustafa

**Apologies:** None

**Minute Taker:** Isaac Lynn

## Agenda

1. Conduct a preliminary client interview run-through.
2. Review UML sequence-diagram standards and Customer–Ops system-boundary notation.
3. Review the alignment of the draft use cases with the developing functional requirements.

## Discussion Summary

- The team discussed responsibility boundaries relating to pricing, overdue-fee calculations and the information exchanged between ChargeMate Customer and ChargeMate Ops.
- The customer-facing pricing and active-rental information requirements were reviewed.

## Decisions

- Interaction diagrams would distinguish the Customer, ChargeMate Customer and ChargeMate Ops as separate participants.
- The team agreed on a consistent approach for representing required and optional interaction flows.

### Actions

- Finalise the stakeholder interview questions.
- Prepare initial diagram drafts in Draw.io.
- Continue updating the SRS structure.
- Develop alternative flows for UC-03 based on the identified error conditions.

### Next Meeting

1 September 2026

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## Meeting Minutes — 1 September 2026

**Date:** 1 September 2026

**Meeting Place:** Macquarie University Library

**Present:** Harrison McLuckie, Cooper Went, Isaac Lynn, Yousef Mustafa

**Apologies:** None

**Minute Taker:** Isaac Lynn

### Agenda

1. Review the SRS baseline structure and confirm the ChargeMate Customer vision and scope.
2. Review stakeholder information relating to station information, availability and error handling.

### Discussion Summary

- The initial SRS repository structure was reviewed.
- Team A reviewed the developing stakeholder information and requirements.
- Exact station availability, event-based availability updates and cross-station return behaviour were discussed.

### Decisions

- The SRS scope and Customer–Ops system boundaries were retained as the baseline for further requirements development.
- The customer interface would present relevant station availability and return information.

### Actions

- Review the remaining stakeholder questions regarding pricing, fee calculation and guest rental.
- Continue converting stakeholder information into formal functional requirements.
- Review the non-functional requirements against the stakeholder clarification evidence.

## Next Meeting

7 September 2026

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## Stakeholder Requirements Clarification — 7 September 2026

**Date:** 7 September 2026

**Meeting Place:** Macquarie University Library

**Present:** Harrison McLuckie, Cooper Went, Isaac Lynn, Yousef Mustafa

**Apologies:** None

**Minute Taker:** Isaac Lynn

### Agenda

1. Clarify the information exchanged between ChargeMate Customer and ChargeMate Ops.
2. Confirm rental, return and error-handling responsibilities.

### Discussion Summary

- Team B confirmed that ChargeMate Customer will receive station location, exact available power-bank counts, charge status, available return slots and station operational status.
- For rentals, ChargeMate Customer sends the power-bank ID, station ID and required rental details to ChargeMate Ops.
- ChargeMate Ops confirms whether the rental and power-bank release were successful.
- For returns, the power-bank ID and return-station information are exchanged with Ops.
- Full return stations, failed releases and failed returns require clear customer-facing messages and appropriate next steps.
- Active-rental information includes start time, elapsed time, current cost, rental status and available return locations.

### Decisions

- Exact station availability information will be displayed to customers.
- Cross-station returns are permitted.
- Guest/minimal-registration rentals are permitted.
- Customer and Ops responsibilities will follow the confirmed stakeholder clarification.

### Actions

- Update the SRS requirements using the confirmed stakeholder responses.
- Update the Overall Description and context information where required.
- Review functional and non-functional requirements against the stakeholder evidence.

For the complete stakeholder questions, responses and requirement mappings, see **Appendix A — Stakeholder Interaction Log**.

## Next Meeting

12 September 2026

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## Meeting Minutes — 12 September 2026

**Date:** 12 September 2026

**Meeting Place:** Zoom

**Present:** Harrison McLuckie, Cooper Went, Isaac Lynn, Yousef Mustafa

**Apologies:** None

**Minute Taker:** Isaac Lynn

### Agenda

1. Review the functional and non-functional requirement fit criteria.
2. Review pricing, rental-duration and late-fee ownership assumptions.
3. Finalise the use case diagram and the three primary use cases.

### Discussion Summary

- Fit criteria for FR-01 through FR-18 and NFR-01 through NFR-05 were reviewed.
- The ChargeMate Customer use case diagram was reviewed.
- UC-01 — Rent a Power Bank, UC-02 — Return a Power Bank and UC-03 — Handle Rental/Return Problem were confirmed as the three use cases for detailed analysis.

### Decisions

- The SRS baseline was updated with consistent vision, scope, requirements and ChargeMate Ops boundaries.
- The three selected use cases would be completed with descriptions and interaction diagrams.

### Actions

- Finalise the descriptions and interaction diagrams for UC-01, UC-02 and UC-03.
- Perform the final document assembly and formatting review.
- Prepare the final Assignment 1 PDF and handover video.

## Appendix C — References and Source Documents

| Source                                               | Use in Project                                                                                                                             |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ChargeMate Customer Project Brief                    | Established the initial problem, customer-facing system scope, project goals and constraints.                                              |
| COMP2050 Assignment 1 SRS Template                   | Provided the required structure for the SRS, Group Activity Record and Appendices.                                                         |
| Team B Requirements Clarification — 7 September 2026 | Used to refine Customer–Ops information exchange, rental and return behaviour, pricing responsibilities, guest rentals and error handling. |

| Source                                         | Use in Project                                                                                       |
|------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Team A Meeting Minutes — August–September 2026 | Provided evidence of team requirements analysis, decisions, task allocation and project development. |
| GitHub Repository History                      | Provided version history and evidence of document development and individual contributions.          |
| LifeSaver Power                                | Reviewed as an existing power-bank rental service during requirements elicitation and comparison.    |
| Ezycharge                                      | Reviewed as an Australian power-bank rental service, including its cross-location return model.      |

## Appendix D — Third-Party Resources

### OpenAI ChatGPT

- **Provider:** OpenAI
- **Tool:** ChatGPT
- **Use:** Used as an editorial and review aid to improve wording, clarify selected requirements, check consistency between SRS sections, and provide feedback on use-case and diagram structure.
- **Scope:** The tool supported refinement and quality checking of existing project work. Stakeholder responses, meeting evidence and individual team contributions were based on the team's own project records.

### diagrams.net / Draw.io

- **Provider:** JGraph Ltd
- **Tool:** diagrams.net / Draw.io
- **Use:** Used to create and edit the ChargeMate Customer context diagram and use case diagram.

### Mermaid.js.org

- **Provider:** Open Source
- **Tool:** Mermaid.js.org
- **Use:** Used to create Context diagram for the Return

## References

- LifeSaver Power - <https://www.lifesaverpower.com/>
- Ezycharge - <https://www.ezycharge.com.au/>